

Integration by parts - 3.1

Product rule: $\frac{d}{dx} (f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$

Integrating gives $f(x)g(x) = \int f'(x)g(x) + f(x)g'(x) dx$

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx$$

Set $u = f(x)$ $du = f'(x)$
 $v = g(x)$ $dv = g'(x)$

$$\boxed{\int u dv = uv - \int v du}$$

Integration by parts

Example : Find $\int x \sin(x) dx$.

$$\begin{array}{l} u = x \\ dv = \sin(x) \end{array} \Rightarrow \begin{array}{l} du = 1 \\ v = -\cos(x) \end{array}$$

$$\begin{aligned} \int x \sin(x) dx &= -x \cos(x) - \int -\cos(x) dx \\ &= -x \cos(x) + \int \cos(x) dx \\ &= -x \cos(x) + \sin(x) + C \end{aligned}$$

Example: Find $\int x e^{2x} dx$.

$$\begin{aligned} u &= x & du &= 1 \\ dv &= e^{2x} & \Rightarrow & v = \frac{1}{2} e^{2x} \end{aligned}$$

$$\begin{aligned} \int x e^{2x} dx &= \frac{x}{2} e^{2x} - \int \frac{1}{2} e^{2x} dx \\ &= \frac{x}{2} e^{2x} - \frac{1}{4} e^{2x} + C \end{aligned}$$

check $\frac{d}{dx} \left(\frac{x}{2} e^{2x} - \frac{1}{4} e^{2x} + C \right)$

$$= \frac{1}{2} e^{2x} + x e^{2x} - \frac{1}{2} e^{2x} = x e^{2x} \quad \checkmark$$

Example: Find $\int \frac{\ln(x)}{x^3} dx$

$$= \int x^{-3} \ln(x) dx \quad \begin{array}{l} u = \ln(x) \\ dv = x^{-3} \end{array} \Rightarrow \begin{array}{l} du = \frac{1}{x} \\ v = \frac{x^{-2}}{-2} = -\frac{1}{2} x^{-2} \end{array}$$

$$= -\frac{\ln(x)}{2} x^{-2} - \int -\frac{1}{2} x^{-2} \cdot \frac{1}{x} dx$$

$$= -\frac{\ln(x)}{2x^2} + \frac{1}{2} \int x^{-3} dx = -\frac{\ln(x)}{2x^2} + \frac{1}{2} \cdot -\frac{1}{2} x^{-2} + C$$

$$= -\frac{\ln(x)}{2x^2} - \frac{1}{4x^2} + C = \frac{-2\ln(x) - 1}{4x^2} + C$$

Example: Find $\int \ln(x) dx = \int 1 \cdot \ln(x) dx$

$$\begin{array}{l} u = \ln(x) \\ dv = 1 \end{array} \Rightarrow \begin{array}{l} du = \frac{1}{x} \\ v = x \end{array}$$

$$\int \ln(x) dx = x \ln(x) - \int x \cdot \frac{1}{x} dx$$

$$= x \ln(x) - \int 1 dx = x \ln(x) - x + C$$

$$\boxed{\int \ln(x) dx = x \ln(x) - x + C}$$